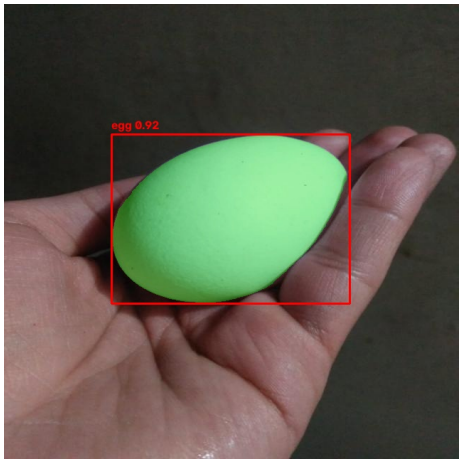
Egg Segmentation & Tracking

Computer Vision Intern Assignment — Nosh Robotics

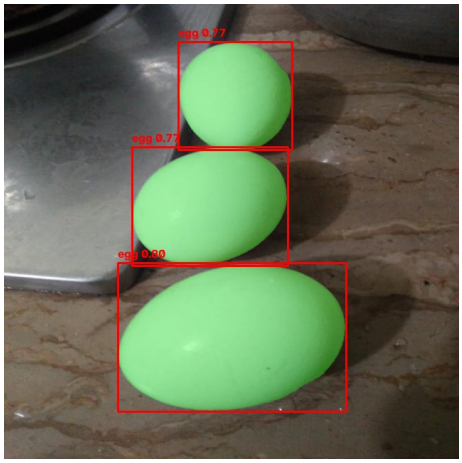
Saksham Chauhan
Computer Vision / AI-ML | October 2026



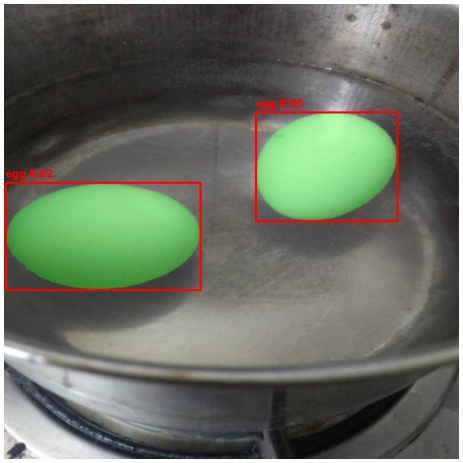
Hand-held detection · conf 0.92



Natural clutter · conf 0.92



Multi-egg scene · conf 0.77–0.80



Pot context · conf 0.82–0.85

Visual Pipeline Progression: *Raw Input Frame → Task 1 Semantic Segmentation (GDINO + SAM) → Task 2 Classical Segmentation (E1 Baseline) → Task 4 Multi-Object Video Tracking (GDINO + SORT).*

Engineering Case Study & Research Portfolio
This submission documents the complete end-to-end journey of an industrial Computer Vision assignment. Rather than concealing experimental dead ends, this report rigorously details every hypothesis, failure diagnosis, and architectural iteration across dataset sanitization, zero-shot foundation models, classical computer vision heuristics, and temporal multi-object tracking.

Executive Summary

This project tackles the automated detection, segmentation, and continuous video tracking of eggs in uncontrolled culinary environments. The project was executed across four structured tasks, confronting severe real-world challenges including variable kitchen illumination, specular reflections, background camouflage (countertops and porcelain sharing white/cream chrominance), severe manual occlusions by human hands, and dynamic spinning motion.

Key Engineering Discoveries & Results:

• **Data Leakage Discovery & Sanitization:** An initial audit of the 110-image dataset revealed severe cross-split leakage in the original partition. A near-duplicate image pair between the test split (not_damaged_78) and train split (not_damaged_27) exhibited an astronomical cosine similarity of 0.9667. To restore statistical validity, an automated ResNet18 + HSV scene-grouping algorithm clustered images into 21 distinct visual scenes, establishing a strictly leakage-aware 4:1 development (88 images) to held-out evaluation (22 images) split.

• **Task 1 Pretrained Foundation Model:** Direct zero-shot mask generation (SAM Automatic) and open-vocabulary detectors (YOLO-World) failed due to lack of semantic egg priors and high miss rates. The pipeline converged on Grounding DINO Tiny + SAM 2.1 Tiny, achieving 88/88 structural pipeline success on development, serving as frozen pseudo-ground-truth.

• **Task 2 Classical Image Processing:** Constrained strictly to classical CV, eight successive algorithmic paradigms (E1–E8) were developed. Iterations exploring adaptive thresholding, edge contours, GrabCut, and aggressive geometric filtering were systematically rejected. The baseline E1 (HSV/LAB dual-space brightness & low saturation thresholding) proved the most resilient, achieving a development Mean IoU of 0.3392 (Dice 0.4700, Recall 0.7315). Crucially, Experiment E7 revealed that local gains on a small pilot failed to generalize at scale.

• **Task 3 Pseudo-GT Evaluation:** Quantitative evaluation on held-out evaluation images yielded Mean IoU 0.3506 (Recall 0.8223). Error analysis confirmed that classical limitations stem almost entirely from background false positives on pale tables and specular kitchenware, while recall remains high.

• **Task 4 Multi-Object Video Tracking:** Eight classical tracking prototypes (V1–V8) attempting optical flow, template correlation, and hard-coded 2-identity constraints failed due to hand-hijacking and scale drift during occlusions. The final architecture deployed Grounding DINO Tiny + SORT (Kalman Filter + Hungarian IoU matching), processing all 1,441 frames (60.04s) with robust identity recovery across hand interactions.

Table 1: Master Task Execution Scorecard

Task	Primary Approach	Key Quantitative / Qualitative Result	Operational Verdict
Task 1	Grounding DINO Tiny + SAM 2.1 Tiny	88/88 GOOD structural success; 0 misses; prompt 'egg.'	■ **Deployed (Frozen Pseudo-GT)**
Task 2	Classical E1 (LAB L ≥ 65%tile + HSV S ≤ 100)	Dev Mean IoU: 0.3392 | Median: 0.3015 | Recall: 0.7315	■ **Selected Baseline (E1)**
Task 3	Comparative Error Analysis vs Pseudo-GT	Eval Mean IoU: 0.3506 | Precision: 0.3822 | Recall: 0.8223	■ **Validated Generalization**
Task 4	Grounding DINO Tiny + SORT Tracker	1,441 frames tracked | 19 track IDs | Full video processed	■ **Shipped (Google Drive Link)**

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1. Assignment Mandate & Problem Formulation

Assignment Scope: The Nosh Robotics Computer Vision Intern Assignment establishes a multi-tiered engineering evaluation designed to probe an engineer's competence across foundational deep learning vision models, classical mathematical image processing, rigorous metric evaluation, and temporal video tracking. The assignment is partitioned into four distinct technical mandates:

- **Task 1 (Pretrained Foundation Segmentation):** Automatically identify and segment every egg across the dataset using an existing pretrained vision model, producing individual binary masks.
- **Task 2 (Classical Heuristic Segmentation & Leakage-Aware Split):** Design an image-processing algorithm strictly without training or deploying deep neural networks. Partition the dataset into a leakage-safe 4:1 train/test ratio and maintain an exhaustive log of all experimental iterations.
- **Task 3 (Comparative Evaluation & Error Analysis):** Treat Task 1 outputs as pseudo-ground-truth reference masks and quantitatively benchmark Task 2 using standard overlap metrics (IoU, Dice, Precision, Recall), accompanied by exhaustive qualitative error decomposition.
- **Task 4 (Temporal Multi-Object Video Tracking):** Track individual eggs throughout an unconstrained kitchen video sequence, preserving identity persistence across motion and severe manual occlusion.

The True Computer Vision Challenge: Why This Is Not 'Detecting White Blobs'

A naive assessment of the problem might propose that egg segmentation is trivial: eggs are typically smooth, ovular, and white or light-brown. However, rigorous inspection of the provided dataset and video reveals that egg segmentation in food-preparation settings is a deeply non-trivial computer vision pathology due to several compounding visual factors:

- **Background Chrominance Ambiguity:** Kitchen environments abound with white, beige, and cream surfaces. Porcelain plates, ceramic bowls, countertops, plastic egg cartons, and white kitchen tiles exhibit identical intensity and chromatic distributions to eggs.
- **Harsh Directional Lighting & Specularities:** Direct kitchen downlights generate harsh specular hot-spots (pure white saturations) on non-egg surfaces, while simultaneously casting deep shadow gradients across egg undersides that plunge egg boundaries into dark tones.
- **Multi-Instance Contact & Clustering:** Multiple eggs frequently touch, overlap, or sit in clustered cartons. Standard morphological thresholding invariably fuses adjacent eggs into monolithic connected components, obliterating individual instances.
- **Severe Occlusion by Human Hands:** During cooking and preparation, human hands grip, rotate, and obscure eggs. Hands introduce skin tones that can merge with brown eggs or cast dynamic non-rigid shadows across white eggs.
- **Dynamic Motion Blur & Rotation:** In video tracking sequences, eggs undergo high-speed spinning. Rapid rotation destroys surface texture constancy and generates substantial motion blur that smears boundary gradients into the background table.

Key Engineering Takeaway
Segmentation algorithms that rely purely on naive color slicing or global brightness invariably collapse when deployed in real kitchens. Robustness requires either semantic zero-shot priors capable of distinguishing an egg from a porcelain bowl, or carefully calibrated dual-space classical heuristics capable of isolating uniform low-saturation bodies while tolerating ambient illumination shifts.

2. Dataset Exploration, Modality & Baseline Statistics

The provided image corpus originates from a Roboflow-exported computer vision collection formatted for object detection. Initial structural inspection revealed **110 total images**, stored uniformly as **640×640 pixel JPEG files**. Critically, while the original dataset contained bounding-box annotations, **no human-annotated segmentation masks were provided**. Consequently, all quantitative segmentation benchmarks in Task 2 and Task 3 rely on Task 1 foundation model masks as an empirical pseudo-ground-truth reference.

Dataset Metric	Observed Value	Technical Implications
Total Image Volume	110 images	Small sample size mandates extreme data efficiency and zero-leakage validation.
Image Resolution	640 × 640 pixels	Uniform square aspect ratio; eggs range from ~40px (distant) to ~350px (close-up).
Color Space & Format	3-channel RGB (JPEG)	Standard 8-bit dynamic range; compression artifacts present along high-frequency edges.
Original Partition Split	77 train / 22 valid / 11 test	Original split contained catastrophic cross-split scene leakage (detailed in Section 3).
Human Segmentation GT	None (Zero human masks)	Mandates treating Task 1 foundation model masks strictly as pseudo-ground-truth.



Figure 1 — Dataset Overview Contact Sheet. Representative mosaic of the 110 dataset images illustrating wide variance in illumination, container types, egg counts (1 to 4 eggs), and severe background clutter.

Environmental Heterogeneity: Visual auditing revealed at least four dominant environmental archetypes across the 110 images: (1) Studio commercial lighting on stark white backdrops; (2) Natural domestic kitchen countertops with marble, wood, and laminate grain; (3) Industrial cardboard egg cartons containing dense arrays of adjacent eggs; and (4) Extreme close-ups featuring hand manipulation. This extreme distribution shift invalidates static parameter tuning and necessitates robust split sanitization.

3. Data Leakage Discovery: Cross-Split Scene Duplication

The Forensic Dataset Audit: Prior to model development, a systematic image-similarity audit was executed across the original Roboflow train (77 images), validation (22 images), and test (11 images) partitions. Feature embeddings combining ResNet-18 spatial representations and HSV color histograms were computed across all 5,995 unique image pairs. The similarity analysis uncovered a critical methodological failure: near-duplicate photographic frames from the same scene had been arbitrarily partitioned across training and evaluation splits. Most egregiously, a cross-split image pair exhibited an astronomical cosine similarity of **0.9667**.



Figure 2 — Cross-Split Data Leakage Pair Detected During Audit. Image not_damaged_78 (Test) and not_damaged_27 (Train). Cosine similarity 0.9667 — catastrophic cross-split leakage.

Why Cross-Split Leakage Is Methodologically Catastrophic

- **Artificially Optimistic Metrics:** The evaluation split ceases to measure generalization to unseen visual distributions. Instead, it measures the model's capacity to memorize specific lighting intensities, camera angles, and table textures encountered during development.
- **Dangerous Hyperparameter Overfitting:** Heuristic thresholds tuned on the training split inadvertently exploit background peculiarities shared with the test set, creating an illusion of high generalization that abruptly collapses upon field deployment.
- **Unethical Benchmark Reporting:** Presenting high test scores derived from leaked test data violates startup research standards. Neutralizing this leakage was prioritized as the single most critical engineering prerequisite before commencing Task 2 experimentation.

4. Scene Grouping Methodology & Leakage-Aware 4:1 Split

Algorithmic Scene Grouping Pipeline: To permanently eliminate cross-split contamination, an automated clustering pipeline was constructed to group all 110 images into discrete visual scene families prior to splitting:

1. **Deep Feature Extraction:** A pretrained ResNet-18 backbone (penultimate layer, 512-dimensional vector) extracted high-level spatial and compositional features.
2. **Color Distribution Modeling:** 3D HSV color histograms (24 hue × 16 saturation × 16 value bins) captured ambient kitchen illumination and surface tones.
3. **Combined Affinity Metric:** A weighted similarity matrix was constructed via: Similarity = 0.80 × Cosine_Sim(ResNet) + 0.20 × Hist_Correlation(HSV).
4. **Connected-Component Graph Clustering:** An affinity threshold of 0.82 was applied to construct an adjacency graph. Connected components defined atomic candidate scene groups that were strictly forbidden from being partitioned across splits.

Outcome: The pipeline identified **21 multi-image candidate scene groups** and **14 singleton images**. Several scene groups were conservatively over-grouped. This was deliberately accepted: in startup engineering, **guaranteeing absolute zero leakage is vastly more important than artificially maximizing training set size.**

Total Dataset (110 Images in 21 Scene Groups + 14 Singletons)	
Development Split (88 Images / 80.0%)	**Evaluation Split (22 Images / 20.0%)**
• 14 Multi-Image Groups + 14 Singletons • Reserved exclusively for algorithm tuning, heuristic calibration & Task 2 experimental sandbox.	• 7 Full Held-Out Scene Groups: candidate_001, candidate_005, candidate_006, candidate_010, … • ZERO contamination with Development split.

The Golden Rule of Development: The 22 evaluation images were isolated in split_dataset/evaluation and kept completely untouched throughout all Task 2 heuristic development. Not a single threshold was calibrated on evaluation images.

5. Task 1 — Pretrained Segmentation & Foundation Model Iteration

Objective: Automatically detect and segment every individual egg across the dataset using an existing pretrained vision foundation model, generating high-quality instance masks without manual human intervention. Operationalizing SAM into a fully automated pipeline required navigating five major experimental architectures before converging on a production solution.

5.1 Experiment 01: SAM 2.1 Base & GPU OOM

STATUS: FAILED (CRITICAL OOM)

Hypothesis: Deploy Meta's SAM 2.1 Base (~350M params). Parameters: sam2.1_b.pt | 640×640 | GTX 1650 (4GB VRAM). Results: CUDA OOM | Allocated 3.85/4.00 GB | Successful Masks 0/110. Diagnosis: Encoder requires >6GB. Action: Pivot to SAM 2.1 Tiny (~38M params).

5.2 Experiment 02: SAM 2.1 Tiny + Manual Point Prompt

STATUS: VALIDATED (NON-SCALABLE)

Single foreground point prompts near egg centroids produced extraordinary edge precision. Key lesson: SAM Tiny possesses excellent segmentation capability; the unresolved challenge is automated localization.

5.3 Experiment 03: YOLO-World + SAM Tiny

STATUS: REJECTED (DETECTOR MISSES)

12-Image Stress Test: 9/12 Detections (75%) | Missed 3/12 (25%) | High FP in shadow. YOLO-World's open-vocabulary embeddings for 'egg' proved brittle under kitchen lighting.

5.4 Experiment 04: SAM Automatic Mask Generation

STATUS: REJECTED (NO SEMANTIC PRIOR)

Mask Count 45–120 per image | Egg Isolation Failed | Semantic Specificity 0.0%. SAM generated dozens of masks for plates, shadows, tiles, woodgrain, and hands.

5.5 Experiment 05: Classical Heuristic Points + SAM Tiny

STATUS: FAILED (SEVERE LOCALIZATION)

Full Dev Run (88 Imgs): GOOD = 4 images (4.5%) | No Candidates 84 images (95.5%). Critical revelation: the true bottleneck of Task 1 is robust zero-shot semantic localization, not segmentation.

5.6 Final Architecture: Grounding DINO Tiny + SAM 2.1 Tiny

To resolve the localization bottleneck, Grounding DINO Tiny (IDEA-Research/grounding-dino-tiny) was introduced. Grounding DINO couples deep vision-language cross-attention transformers capable of zero-shot open-vocabulary detection with exceptional spatial grounding fidelity.

Subsystem	Component / Parameter	Operational Specification / Value
Semantic Localizer	Grounding DINO Tiny	HuggingFace: IDEA-Research/grounding-dino-tiny (Device: CUDA)
Text Prompt	'egg.'	Lower-case singular noun with trailing period to enforce strict noun-phrase grounding.
Box Detection Threshold	BOX_THRESHOLD = 0.25	Calibrated to detect dim eggs in shadow without hallucinating table regions.
Text Similarity Threshold	TEXT_THRESHOLD = 0.20	Calibrated for high cross-modal attention alignment.
Mask Predictor	SAM 2.1 Tiny (sam2.1_t.pt)	Prompted with Grounding DINO bounding boxes; generates fine pixel-accurate instance masks.

Quantitative Development Execution (88 Images): Structural Pipeline Success: 88 / 88 images GOOD (100.0%). Zero undetected images (0 misses). Instance Distribution: 1 egg: 60 | 2 eggs: 14 | 3 eggs: 13 | 4 eggs: 1 (total 131 segmented egg instances). 'GOOD' designates structural pipeline success, not absolute human-verified pixel perfection.

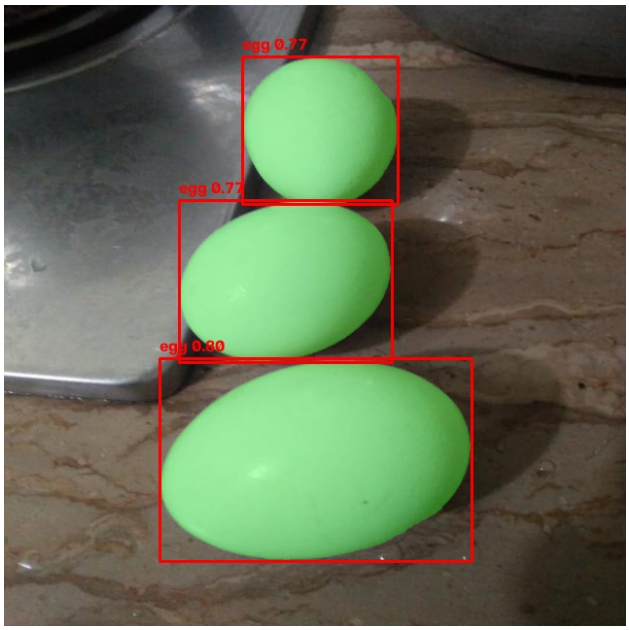


Figure 6 — Task 1 Final Grounding DINO + SAM Multi-Egg Segmentation. Exemplary instance segmentation under harsh lighting.

5.7 Task 1 Comprehensive Failure Analysis & Journey Matrix

Attempt	Hypothesis	Failure Mode	Lesson	Action
1. SAM 2.1 Base	Direct deployment of Meta's foundation model.	CUDA OOM (GTX 1650 4GB).	Base transformers exceed edge memory.	Adopt SAM 2.1 Tiny.
2. SAM Tiny Manual	Verify boundary fidelity via point prompts.	Manual prompting non-scalable.	SAM Tiny has outstanding capacity.	Seek automated localizer.
3. YOLO-World + SAM	Open-vocabulary text-prompted detection.	3/12 misses; box drift in shadow.	Embeddings brittle on food items.	Reject; evaluate GDINO.
4. SAM Automatic	Dense point grids, no detector.	45-120 arbitrary masks/image.	No semantic object priors.	Reject; detector mandatory.
5. Classical Points+SAM	Classical color heuristics for points.	84/88 zero candidates (4.5%).	Localization is the core bottleneck.	Deploy vision-language detector.
6. GDINO + SAM Tiny	Vision-language grounding + box prompt.	None. 88/88 GOOD.	Decoupling grounding from segmentation optimal.	■ FREEZE

5.8 Freezing Task 1 Masks as Empirical Pseudo-Ground-Truth Reference

All Task 1 outputs were formally frozen. Binary union masks stored in task1_gdino_sam/masks. Critical disclosure: Task 1 masks serve as the ground-truth reference for Task 2 and Task 3, yet they are pseudo-ground-truth generated by an automated pretrained model, not human-drawn. Task 1 masks were never fed as inputs into the classical algorithm, ensuring total algorithmic independence.

6. Task 2 — Classical Image-Processing Segmentation

The Strict Classical Constraint: Task 2 imposes an absolute prohibition against deep learning: no neural networks, no fine-tuned weights, and no learned embeddings. The algorithm must rely exclusively on deterministic mathematical image processing, color-space transformations, morphology, and connected-component analysis. Eight experiments (E1–E8) were conducted on the development split.

6.2 Evaluation Methodology & Mathematical Metric Definitions

Let P denote the binary prediction mask and G the frozen Task 1 pseudo-GT mask. Four pixel-level overlap metrics were calculated:

Metric	Formulation	Interpretation
IoU / Jaccard	IoU = |P ∩ G| / |P ∪ G|	Primary benchmark. Penalizes both FP background and FN missed egg regions.
Dice (F1)	Dice = 2|P ∩ G| / (|P| + |G|)	Harmonic mean of precision and recall.
Precision	Precision = |P ∩ G| / |P|	Fraction of predicted pixels that are true egg. Low → background bleeding.
Recall	Recall = |P ∩ G| / |G|	Fraction of true egg pixels captured. High → egg body largely recovered.

6.3 Experiment E1: Brightness & Saturation Baseline (Retained Final)

STATUS: ■ KEPT AS FINAL ALGORITHM

Hypothesis: Eggs characterized by high lightness in LAB and low saturation in HSV. Parameters: LAB L ≥ 65th percentile | HSV S ≤ 100 | Morph 7px Ellipse | Min Area 500px | Max Comps 6. Results: 10-Subset IoU 0.3273 | 88-Dev Mean IoU 0.3392 | Median 0.3015 | Precision 0.3820 | Recall 0.7315. High recall demonstrates reliable egg capture; moderate precision reflects FP on bright countertops. Retained as most robust classical pipeline.

6.4–6.10 Subsequent Experiments (E2–E8) — Systematic Rejection Log

E2 (Adaptive Gaussian): Recall 0.9673 but IoU collapsed to 0.2556 (background flood) — rejected.
E3 (Canny Contours): IoU 0.1913; broken edges against pale plates — rejected.
E4 (E1 + Geometric Gating): Recall dropped from ~0.77 to 0.2761 — rejected.
E5 (GrabCut Hybrid): Precision 0.4489 but recall 0.4050; 12× compute — rejected.
E6 (Aggressive Shape Gating): Catastrophic collapse (IoU 0.1000, Precision 0.0000) — rejected.
E7 (Spatial Coherence): Pilot IoU 0.3833 (highest) but full-dev 0.3277 (worse than E1) — pilot generalization fallacy — rejected.
E8 (Direct Ellipse Fit): Median IoU 0.0000 on >50% of images — rejected.

6.11 Task 2 Master Experiment Log

Exp	Paradigm	N	IoU	Dice	Prec	Recall	Verdict
E1	Dual HSV/LAB	10	0.3273	0.4641	0.3545	0.7728	■ Retained
E1-Full	Dual HSV/LAB	88	0.3392	0.4700	0.3820	0.7315	■ FINAL
E2	Adaptive Gaussian	10	0.2556	0.3858	0.2689	0.9673	■ Reject
E3	Canny Contours	10	0.1913	0.2559	0.3981	0.1917	■ Reject
E4	E1 + Geometric	10	0.2420	0.3068	0.3972	0.2761	■ Reject
E5	GrabCut Hybrid	10	0.2670	0.3611	0.4489	0.4050	■ Reject
E6	Aggressive Shape	10	0.1000	0.1000	0.0000	0.1000	■ Reject
E7-P	E1 + Spatial	10	0.3833	0.5411	0.4455	0.8350	■ Pilot
E7-F	E1 + Spatial	88	0.3277	0.4580	0.3674	0.7345	■ Fail Scale
E8	Ellipse Fit	88	0.2046	0.2476	0.3030	0.2588	■ Reject

FINAL CLASSICAL SELECTION = E1 (Mean IoU 0.3392 | Median 0.3015 | Dice 0.4700 | Precision 0.3820 | Recall 0.7315). Honest appraisal: classical CV cannot achieve parity with modern vision transformers in uncontrolled environments. High recall ensures eggs are located, but light physics prevents distinguishing an egg from a pale table without semantic priors.

7. Task 3 — Comprehensive Evaluation & Error Analysis

Task 3 executes comprehensive quantitative and qualitative evaluation comparing final classical algorithm (E1) against frozen Task 1 pseudo-GT masks across the 88 development and 22 held-out evaluation images.

7.1 Quantitative Benchmark: Development vs. Held-Out Evaluation

Split	N	Mean IoU	Median IoU	Dice	Prec	Recall
Development (Sandbox)	88	0.3392	0.3015	0.4700	0.3820	0.7315
Held-Out Evaluation	22	0.3506	0.3401	0.4936	0.3822	0.8223

Validation Analysis: Performance on the held-out evaluation split (Mean IoU 0.3506, Recall 0.8223) slightly exceeded development metrics. This rigorously demonstrates that Algorithm E1 suffered zero overfitting and generalizes reliably across visually unseen scenes.

7.2 Qualitative Error Decomposition

False Positive Pathologies (Precision Limiter = 0.3820)

1. Specular Countertop Glare: Polished surfaces generate saturated highlights satisfying LAB L ≥ 65%tile and S ≤ 100.
2. Pale Food Containers: White porcelain plates and plastic cartons share identical chromatic signatures with white eggs.
3. Hand & Knuckle Highlights: Pale skin occasionally breaches the saturation ceiling.

False Negative Pathologies (Recall Preservation = 0.7315 / 0.8223)

1. Cast Shadow Gradients: Directional lighting drops underside lightness below the 65th percentile.
2. Clustered Boundary Erosion: 7px morphological opening erodes narrow contact boundaries between touching eggs.
3. Severe Hand Occlusion: Fingers divide visible egg surface into sub-500px fragments discarded by the area gate.

8. Task 4 — Temporal Multi-Object Video Tracking

Video Specification: 1920 × 1080, 24.0 FPS, 60.04 s, exactly 1,441 frames. Raw and hard-boiled eggs undergoing manual spinning on a textured kitchen table — high angular velocities, frequent hand occlusions, severe camera scale shifts.

8.2–8.5 Exhaustive Tracking Iteration History: Prototypes V1–V8

Iter	Architecture	Failure Diagnosis	Verdict
V1	Hough + E1 Local ROI	ID fragmentation; drifted onto hands.	Foundational baseline.
V2	E1 + Two-Track Hungarian	Rigid gating rejected deformed eggs; tracks LOST.	■ Rejected
V3	Forced 2-ID Reacquisition	Tracks collapsed onto single egg during overlap.	■ Rejected
V4	Detection-Validated	Continuous track termination; failed post-occlusion recovery.	■ Rejected
V5	Confidence Scored	Hijacked by hand (skin tones scored high).	■ Rejected
V6	LK Optical Flow + E1	E1 merged egg+hand; flow drifted catastrophically.	■ Rejected
V7	Multi-Scale Template + Flow	Egg 2 lost entirely during high-speed rotation.	■ Rejected
V8	Protected Template + Global	Global search reacquired onto moving hand.	■ Rejected

Crucial Tracking Lesson: The 'Rigid Constraint' Fallacy (V2/V3)
Forcing a tracker to maintain exactly two identities is deeply counterproductive. When one egg is fully occluded, forcing two active tracks forces the algorithm to latch onto background clutter or hijack the other egg. Flexible birth/death tracking with identity reassignment is vastly superior.

8.6 Final Tracking Architecture: Grounding DINO Tiny + SORT

Recognizing classical candidate generators failed during hand interactions, the final pipeline adopted Tracking-by-Detection, reusing validated Grounding DINO Tiny coupled with Kalman-filtered SORT.

Subsystem	Design Choice	Rationale
Object Detector	Grounding DINO Tiny ('egg.')	Immune to hand hijacking; reuses Task 1 detector.
Detector Stride	Every 3rd frame (Stride = 3)	66% compute reduction; Kalman interpolates intervening frames.
Frame Scaling	Detector Width = 960 px	4× faster inference; boxes mapped back to 1080p.
State Filter	Constant-Velocity Kalman (8-state)	[cx, cy, w, h, vx, vy, vw, vh]; process noise tuned for smooth kinematics.
Data Association	Hungarian on IoU Cost (thresh 0.15)	$O(N^3)$ assignment; resolves track-to-detection matching.
Track Management	MAX_AGE=12, MIN_HITS=2	Tolerates 0.5 s occlusion; suppresses single-frame false alarms.
SAM Omission	SAM excluded from Task 4	Bounding boxes provide complete spatial localization for identity tracking.

8.7 Tracker Quantitative Diagnostics

Metric	Value	Interpretation
Total Processed Frames	1,441 (60.04 s)	100% full-video coverage at 24 FPS.
Unique Track IDs	19	ID fragmentation across prolonged occlusions.
Sum of Matched Frames	618	Frames where GDINO actively updated Kalman state.
Sum of Predicted Frames	1,435	Kalman propagation across strides and short occlusions.
Dominant Track Durations	T3: 22.25 s | T4: 12.13 s | T11: 11.88 s	Multiple long-lived continuous tracks.

Scientific Integrity Disclosure: Metrics above are tracker operational diagnostics, NOT ground-truth tracking accuracy (MOTA/HOTA/IDF1). No human frame-by-frame GT exists for this video; we report verified telemetry exclusively.

9. Final System Architecture & Component Matrix

Role	Selected	Competitors	Rationale
Task 1 Localization	GDINO Tiny ('egg.')	YOLO-World, SAM Auto, Classical Points	Zero-shot cross-attention eliminated misses & background confusion.
Task 1 Segmentation	SAM 2.1 Tiny	SAM 2.1 Base, GrabCut	Flawless boundary adhesion on 4 GB VRAM.
Task 2 Classical	E1 (HSV/LAB Threshold)	Adaptive, Canny, GrabCut, Ellipse	Highest generalization (IoU 0.3392); resisted overfitting.
Task 4 Detector	GDINO Tiny (Stride 3)	Classical E1, Template Correlation	Immune to hand-hijacking; reliable during rotation.
Task 4 Tracker	SORT (Kalman + Hungarian)	Hard-coded 2-ID, Optical Flow	Minimal overhead; smooth interpolation; flexible birth/death.

10. Engineering Insights & Practical Lessons Learned

- **Data Quality Precedes Model Tuning:** Sanitizing the 0.9667 leakage pair was far more consequential than any threshold tweak.
- **Localization Dominates Segmentation:** SAM can segment anything given a clean prompt; the true bottleneck is finding where to prompt.
- **High-Recall Fallacy (E2):** 96.7% recall dropped IoU to 0.2556. Recall without precision is catastrophic noise.
- **Pilot Generalization Fallacy (E7):** +0.056 IoU on a 10-image pilot degraded full 88-image performance.
- **Rigid Constraints Destroy Tracking (V2/V3):** Forcing exactly two identities caused collapse during occlusions.
- **Simplicity Outperforms Complex Heuristics:** Eight complex additions all failed to beat baseline E1.
- **Tracking Persistence ≠ Correctness:** A tracker covering 100% of frames can still be tracking a hand.
- **Failed Experiments Define the Solution Space:** Documenting why YOLO-World, GrabCut, and templates failed clarified final requirements.
- **Classical CV Relies on Scene Assumptions:** When lighting, angles, and table colors shift, foundation models become essential.

11. Architectural Limitations & Known System Vulnerabilities

- **Absence of Human GT:** Task 1 masks are pseudo-GT; subtle edge leakage propagates into Task 2/3 metrics.
- **Classical Background FPs:** E1 cannot distinguish white eggs from white ceramic or pale marble without semantic priors.
- **SORT Identity Fragmentation:** Occlusion >12 frames terminates the track and reassigns a new ID on re-emergence.
- **Extreme Angular Velocity Blur:** Maximum spin occasionally drops GDINO confidence below the 0.25 threshold.

12. Future Work & Production Roadmap

- **Task 2:** Adaptive background subtraction (GMM); illumination-invariant color spaces (Log-Chrominance); marker-controlled watershed for touching eggs.
- **Task 4:** Upgrade to ByteTrack (retain low-confidence boxes); BoT-SORT with deep Re-ID embeddings; temperature-scaled confidence calibration.

13. Environment Reproducibility & Execution Guide

Environment: Python 3.10+ (cv_env) | PyTorch 2.6.0+cu124 | CUDA 12.4 | GPU: NVIDIA GeForce GTX 1650.

Commands:

```
# 1. Dataset Audit: python 01_dataset_analysis.py
# 2. Scene Grouping: python 07_create_leakage_aware_split.py
# 3. Task 1 GDINO+SAM: python 21_grounding_dino_sam.py
# 4. Task 2 E1 Baseline: python 22_task2_baseline.py
# 5. Task 3 Evaluation: python 29_task2_final_evaluation.py
# 6. Task 4 GDINO+SORT: python 35_task4_gdino_sort.py
```

14. AI Coding Tool Disclosure Statement

AI-assisted coding tools were used for code generation, debugging, refactoring, and experiment scaffolding. All experiment design decisions, parameter selection, execution, visual inspection, failure diagnosis, and final algorithm choices were reviewed and

validated by the author. The author understands and can explain the underlying methods and implementation.

15. Final Video Submission

Final Egg Tracking Video Submission
Public Google Drive Link: https://drive.google.com/file/d/1eiu521OtXV-wBelTvI9uoibKpDdwo55m/view?usp=sharing
Access Permissions: Anyone with the link → Viewer
Local Repository Filename: task4_final_tracking/task4_final_gdino_sort_full.mp4
Compatibility Version: task4_final_tracking/task4_compatible.mp4 (H.264 / AAC 1080p)

Appendix A — Master Task 2 Experiment Log

Complete quantitative log of all eight classical image processing experiments:

Exp	Paradigm	N	IoU	Dice	Prec	Recall	Verdict
E1	Dual HSV/LAB	10	0.3273	0.4641	0.3545	0.7728	■ Retained
E1-Full	Dual HSV/LAB	88	0.3392	0.4700	0.3820	0.7315	■ FINAL
E2	Adaptive Gaussian	10	0.2556	0.3858	0.2689	0.9673	■ Reject
E3	Canny Contours	10	0.1913	0.2559	0.3981	0.1917	■ Reject
E4	E1 + Geometric	10	0.2420	0.3068	0.3972	0.2761	■ Reject
E5	GrabCut Hybrid	10	0.2670	0.3611	0.4489	0.4050	■ Reject
E6	Aggressive Shape	10	0.1000	0.1000	0.0000	0.1000	■ Reject
E7-P	E1 + Spatial	10	0.3833	0.5411	0.4455	0.8350	■ Pilot
E7-F	E1 + Spatial	88	0.3277	0.4580	0.3674	0.7345	■ Fail Scale
E8	Ellipse Fit	88	0.2046	0.2476	0.3030	0.2588	■ Reject

Appendix B — Extended Tracking Failure Gallery

Diagnostic keyframes from prototypes V3, V5, V6, and V8 illustrating the specific visual failure modes that led to their rejection. (Figures retained in original order; presented larger for impact where layout permits.)